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# Voyager IQ — Design System

> **Version 1.0 · 3 November 2026** Linh Long Engineering / Rocket EMS · repository `ll-poc` **Source of truth:** `src/theme/` · **Living specimen:** `/design-system`

Everything in this document was read out of a working build. Every number was taken from the source; every picture was captured from the running application at 2× on 3 November 2026. Where a figure came from measuring the screen rather than from a rule, it says so.

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# Part A — Start here

## A1 · What this is

The file someone needs to **build** a new Voyager IQ screen without asking, and the file a reviewer needs to **tell a correct screen from a plausible one**.

It is not a style guide for a drawing tool. It describes a system that is running: the tokens are CSS custom properties, the components are React, and the specimen page renders the real thing.

## A2 · The three layers

| Layer | What it is | Where | May a screen use it? |
|---|---|---|---|
| **1 · Ramps** | The raw palette. 10 blues, 12 greys, 4 reds | `tokens.css` | **No** |
| **2 · Semantic roles** | `surface`, `on-surface`, `brand`, `danger` … named for the job | `tokens.css` | **Yes — only this** |
| **3 · Components** | Button, Field, Grid, Dialog… | `src/ui`, `src/components` | Yes |

A stylesheet saying `background: var(--vy-grey-0)` has hard-coded *white*, and no theme can move it. One saying `var(--vy-surface)` has hard-coded *"the colour a panel is"*, which a theme is allowed to answer differently.

**Layer 1 exists so that layer 2 can be consistent. It is not for use in a component.**




## A3 · Five rules before you build

1. **Is there a token?** If yes, use it. If no, add one — and say in the code why the scale did not already have it.
2. **Does the live system already answer this?** Field names, option lists and behaviour come from the live system unless a customer requirement overrides it. The order of precedence is written down in `precedence.md`.

3. **Both themes, both densities, 820px.** A screen that has only been seen in light mode on a 1920px monitor has been seen once.
4. **A magic number carries a note.** A width that deviates from its role needs a `widthNote` saying what was measured.
5. **Say what is not real.** A derived figure, an unbuilt action and a fake send are each labelled where they appear, not only in a commit message.

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# Part B — Foundations

## B1 · Colour

### B1.1 The ramps *(layer 1 — not for use in components)*

Three ramps and nothing else. A fourth colour in a screen is a bug unless it is a status, and the statuses are in B1.6.



| Token | Value |
| --- | --- |
| `--vy-blue-50` | `#f1f4fa` |
| `--vy-blue-100` | `#dae2f3` |
| `--vy-blue-200` | `#b0c2ea` |
| `--vy-blue-300` | `#7f9ce1` |
| `--vy-blue-400` | `#4973d4` |
| `--vy-blue-500` | `#2754bc` |
| `--vy-blue-600` | `#1a419e` |
| `--vy-blue-700` | `#15347e` |
| `--vy-blue-800` | `#112960` |
| `--vy-blue-900` | `#0d1e44` |
| `--vy-red-50` | `#fdeceb` |
| `--vy-red-500` | `#e23a32` |
| `--vy-red-600` | `#c62d26` |
| `--vy-red-700` | `#9e231d` |
| `--vy-grey-0` | `#ffffff` |
| `--vy-grey-25` | `#fbfcfd` |
| `--vy-grey-50` | `#f5f7fa` |
| `--vy-grey-100` | `#eceff3` |
| `--vy-grey-200` | `#dde2e9` |
| `--vy-grey-300` | `#c3cbd6` |
| `--vy-grey-400` | `#97a2b2` |
| `--vy-grey-500` | `#626d7e` |
| `--vy-grey-600` | `#4d5766` |

| Token | Value |
| --- | --- |
| `--vy-grey-700` | `#363e4b` |
| `--vy-grey-800` | `#232a34` |
| `--vy-grey-900` | `#131820` |

## B1.2 Surfaces — what a thing sits on

`app` is the page · `raised` a dialog or menu · `sunken` an inset below the page · `chrome` the deliberately dark furniture that stays dark in **both** themes.

| Token | Value | Resolves to | Dark |
| --- | --- | --- | --- |
| `--vy-surface-app` | `var(--vy-grey-50)` | `#f5f7fa` | `#0f131a` |
| `--vy-surface` | `var(--vy-grey-0)` | `#ffffff` | `#161c25` |
| `--vy-surface-raised` | `var(--vy-grey-0)` | `#ffffff` | `#1e2531` |
| `--vy-surface-subtle` | `var(--vy-grey-25)` | `#fbfcfd` | `#131922` |
| `--vy-surface-sunken` | `var(--vy-grey-100)` | `#eceff3` | `#0b0e14` |
| `--vy-surface-strong` | `var(--vy-grey-200)` | `#dde2e9` | `#2a323f` |
| `--vy-surface-hover` | `var(--vy-grey-50)` | `#f5f7fa` | `#1e2531` |
| `--vy-surface-selected` | `var(--vy-blue-50)` | `#f1f4fa` | `#1b2a4a` |
| `--vy-surface-chrome` | `var(--vy-grey-900)` | `#131820` | `#1e2531` |
| `--vy-surface-chrome-soft` | `var(--vy-grey-700)` | `#363e4b` | `#2a323f` |

## B1.3 Text colour — six tiers

`on-surface` is body copy; `strong` sits above it for headings and identifiers.

> ⚠️ `on-surface-faint` **measures 2.58:1 on white.** Legitimate only on a dark ground or where the text is decorative — never for anything a user has to read. See E1.

| Token | Value | Resolves to | Dark |
|---|---|---|---|
| `--vy-on-surface-strong` | `var(--vy-grey-900)` | `#131820` | `#f5f8fc` |
| `--vy-on-surface` | `var(--vy-grey-800)` | `#232a34` | `#e9edf3` |
| `--vy-on-surface-soft` | `var(--vy-grey-700)` | `#363e4b` | `#d3dae4` |
| `--vy-on-surface-muted` | `var(--vy-grey-600)` | `#4d5766` | `#aab4c3` |
| `--vy-on-surface-subtle` | `var(--vy-grey-500)` | `#626d7e` | `#8f9aaa` |
| `--vy-on-surface-faint` | `var(--vy-grey-400)` | `#97a2b2` | `#67727f` |
| `--vy-on-surface-invert` | `var(--vy-grey-0)` | `#ffffff` | `#0f131a` |
| `--vy-on-chrome-muted` | `var(--vy-grey-300)` | `#c3cbd6` | `#c3cbd6` |
| `--vy-on-dark-fg` | `#ffffff` | — | *(same)* |
| `--vy-on-dark-secondary` | `var(--vy-grey-300)` | `#c3cbd6` | *(same)* |
| `--vy-on-dark-muted` | `var(--vy-grey-400)` | `#97a2b2` | *(same)* |
| `--vy-on-dark-hairline` | `rgba(255, 255, 255, .07)` | — | *(same)* |
| `--vy-on-dark-hover` | `rgba(255, 255, 255, .05)` | — | *(same)* |
| `--vy-on-dark-active` | `rgba(43, 99, 184, .22)` | — | *(same)* |

## B1.4 Lines

`border` is **structure** — panel edges, table rules — which WCAG 1.4.11 does not cover. `border-control` is the boundary of something you can **operate**, which it does cover.

It is the one value in the system chosen by **measurement** rather than by the ramp. The sweep of 5 September found why: the grey ramp has **no step that conforms**. grey-400 `#97a2b2` is 2.58:1 against white and misses by 0.42; the next step overshoots to 5.24. `--vy-border-control` `#8a94a6` was introduced between them at **3.06:1** — the lightest value that conforms. It governs boundaries only: input, textarea, combobox, checkbox, radio.

| Token | Value | Resolves to | Dark |
|---|---|---|---|
| `--vy-border-control` | `#8a94a6` | — | `#6d7889` |
| `--vy-border-faint` | `var(--vy-grey-100)` | `#eceff3` | `#212936` |
| `--vy-border` | `var(--vy-grey-200)` | `#dde2e9` | `#2a323f` |
| `--vy-border-strong` | `var(--vy-grey-300)` | `#c3cbd6` | `#3b4553` |
| `--vy-border-mid` | `var(--vy-grey-400)` | `#97a2b2` | `#55606f` |
| `--vy-border-heavy` | `var(--vy-grey-500)` | `#626d7e` | `#55606f` |
| `--vy-border-emphasis` | `2px` | — | *(same)* |

## B1.5 Brand and danger

Fill and text are **separate roles**, because in a dark theme they pull opposite ways: a fill must be dark enough for its label, brand text light enough for the page. The dark theme inverts the pair rather than shifting it.

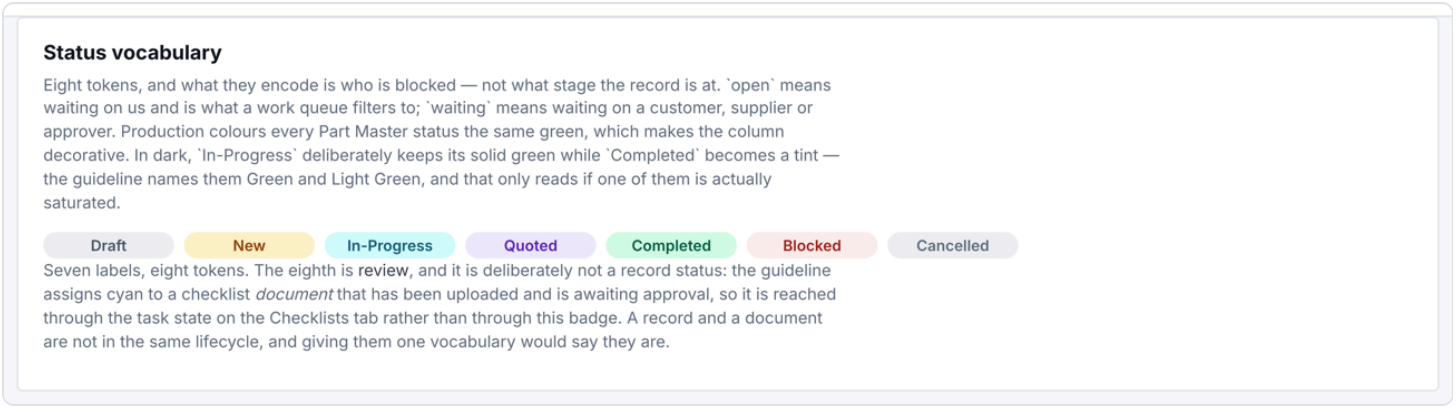
| Token | Value | Resolves to | Dark |
|---|---|---|---|
| `--vy-brand` | `var(--vy-blue-600)` | `#1a419e` | `#7f9ce1` |
| `--vy-brand-hover` | `var(--vy-blue-700)` | `#15347e` | `#b0c2ea` |
| `--vy-brand-active` | `var(--vy-blue-800)` | `#112960` | `#dae2f3` |
| `--vy-brand-text` | `var(--vy-blue-600)` | `#1a419e` | `#7f9ce1` |
| `--vy-brand-text-hover` | `var(--vy-blue-700)` | `#15347e` | `#b0c2ea` |
| `--vy-brand-text-deep` | `var(--vy-blue-800)` | `#112960` | `#dae2f3` |
| `--vy-brand-on` | `var(--vy-grey-0)` | `#ffffff` | `#0f131a` |
| `--vy-brand-subtle` | `var(--vy-blue-50)` | `#f1f4fa` | `#1b2a4a` |
| `--vy-brand-subtle-strong` | `var(--vy-blue-100)` | `#dae2f3` | `#223357` |
| `--vy-brand-light` | `var(--vy-blue-400)` | `#4973d4` | `#7f9ce1` |
| `--vy-brand-soft` | `var(--vy-blue-500)` | `#2754bc` | `#4973d4` |
| `--vy-brand-boundary` | `var(--vy-blue-400)` | `#4973d4` | `#4973d4` |
| `--vy-brand-boundary-soft` | `var(--vy-blue-200)` | `#b0c2ea` | `#2754bc` |
| `--vy-brand-boundary-light` | `var(--vy-blue-300)` | `#7f9ce1` | `#7f9ce1` |
| `--vy-brand-boundary-strong` | `var(--vy-blue-500)` | `#2754bc` | `#4973d4` |

| Token | Value | Resolves to | Dark |
|---|---|---|---|
| `--vy-danger` | `var(--vy-red-600)` | `#c62d26` | `#c62d26` |
| `--vy-danger-on` | `var(--vy-grey-0)` | `#ffffff` | `#ffffff` |
| `--vy-danger-text` | `var(--vy-red-700)` | `#9e231d` | `#f0918a` |
| `--vy-danger-text-deep` | `var(--vy-red-700)` | `#9e231d` | `#f5aba5` |
| `--vy-danger-subtle` | `var(--vy-red-50)` | `#fdeceb` | `#38201e` |
| `--vy-danger-boundary` | `var(--vy-red-500)` | `#e23a32` | `#e0655c` |

## B1.6 Status vocabulary

Eight statuses, each a foreground/background pair.

> **A status is a word and a colour, never a colour alone.** The pair is what makes it legible to someone who cannot separate the hues.



| Token | Value | Resolves to | Dark |
|---|---|---|---|
| `--vy-status-draft-fg` | `var(--vy-grey-600)` | `#4d5766` | `#aab4c3` |
| `--vy-status-draft-bg` | `var(--vy-grey-100)` | `#eceff3` | `#2a323f` |
| `--vy-status-open-fg` | `#92400e` | — | `#eeb796` |
| `--vy-status-open-bg` | `#fef3c7` | — | `#38281e` |
| `--vy-status-progress-fg` | `#155e75` | — | *(same)* |
| `--vy-status-progress-bg` | `#cffafe` | — | *(same)* |
| `--vy-status-done-fg` | `#065f46` | — | `#96eed5` |
| `--vy-status-done-bg` | `#d1fae5` | — | `#1e3831` |
| `--vy-status-waiting-fg` | `#5b21b6` | — | `#b896ee` |
| `--vy-status-waiting-bg` | `#ede9fe` | — | `#281e38` |
| `--vy-status-review-fg` | `#0e7490` | — | `#96dbee` |
| `--vy-status-review-bg` | `#ecfeff` | — | `#1e3338` |
| `--vy-status-blocked-fg` | `var(--vy-red-700)` | `#9e231d` | `#ee9a96` |
| `--vy-status-blocked-bg` | `var(--vy-red-50)` | `#fdeceb` | `#38201e` |
| `--vy-status-cancelled-fg` | `var(--vy-grey-500)` | `#626d7e` | `#aab4c3` |
| `--vy-status-cancelled-bg` | `var(--vy-grey-100)` | `#eceff3` | `#2a323f` |
| `--vy-status-slot-w` | `104px` | — | *(same)* |

## B1.7 Row tones and accents

A tone is a row's **state in the quote**, not a severity. Three of them paint most of a BoM grid, and the legend under the grid names all three.

| Token | Value | Resolves to | Dark |
|---|---|---|---|
| `--vy-tone-excluded-bg` | `var(--vy-grey-100)` | `#eceff3` | `#2a323f` |
| `--vy-tone-excluded-fg` | `var(--vy-grey-500)` | `#626d7e` | `#aab4c3` |
| `--vy-tone-missing-bg` | `var(--vy-red-50)` | `#fdeceb` | `#38201e` |
| `--vy-tone-missing-fg` | `var(--vy-red-700)` | `#9e231d` | `#ee9a96` |
| `--vy-tone-short-bg` | `var(--vy-status-open-bg)` | `#fef3c7` | *(same)* |
| `--vy-tone-covered-bg` | `var(--vy-status-done-bg)` | `#d1fae5` | *(same)* |
| `--vy-tone-known-mfg-bg` | `var(--vy-status-open-bg)` | `#fef3c7` | *(same)* |
| `--vy-tone-pending-bg` | `var(--vy-surface-sunken)` | `#eceff3` | *(same)* |

| Token | Value | Resolves to | Dark |
|---|---|---|---|
| `--vy-accent-positive` | `#047857` | — | `#75f0cd` |
| `--vy-accent-caution` | `#d97706` | — | `#f0b775` |
| `--vy-accent-warning` | `#b45309` | — | `#f0aa75` |
| `--vy-accent-waiting` | `#6d28d9` | — | `#a679ec` |
| `--vy-accent-neutral` | `#4d5766` | — | `#aab4c3` |
| `--vy-accent-bar-w` | `3px` | — | *(same)* |

## B1.8 How dark mode is wired

Three blocks, and **a token must appear in all three** or the two ways of choosing dark disagree:

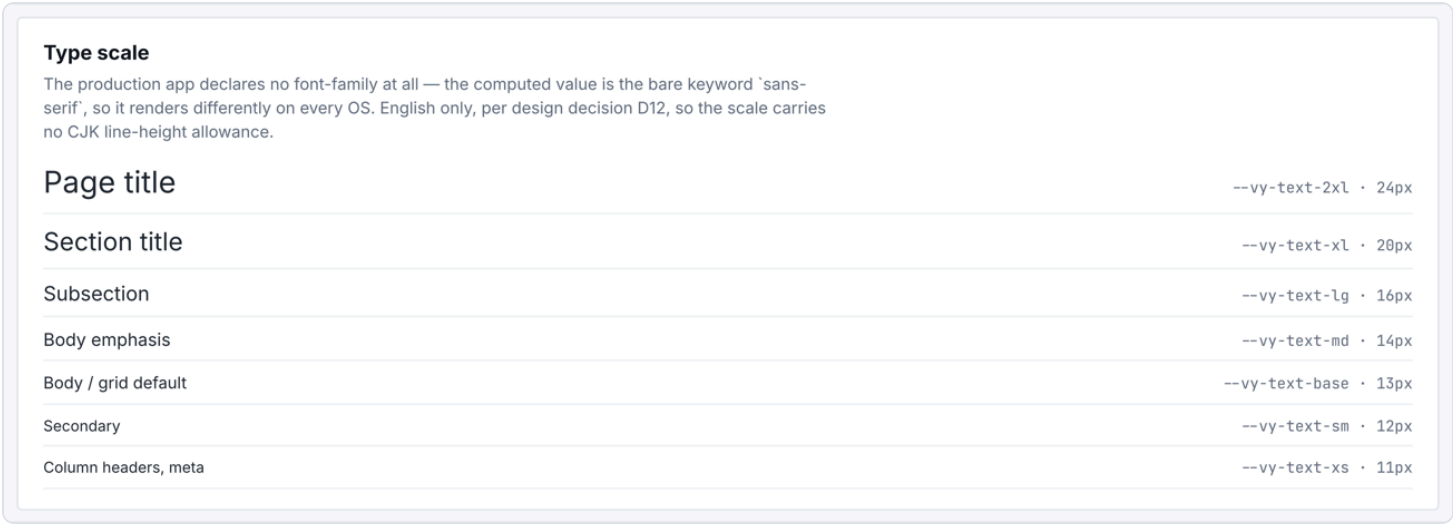
| Block | Who it ser |
| --- | --- |
| `:root { … }` | Light — the default |
| `:root[data-theme="dark"] { … }` | A user who **picked** dar in the user menu |
| `@media (prefers-color-scheme: dark) { :root:not([data-theme="light"]) { … } }` | A user who **OS** is dark |

> This has already failed once: four tokens existed only in the media copy, so a user who picked dark in the menu got grey-400 borders and a near-white priority dot on a near-black ground. **Adding a colour token means editing three places.**

---

## B2 · Typography

Two families. **Inter Variable** for everything; **JetBrains Mono** for identifiers that get read aloud or compared character by character — part numbers, MPNs, error ids. Body text is never mono: the customer's instruction of 13 October, *"do not use monotext, hard to read"*.



| Token | Value |
| --- | --- |
| `--vy-font-sans` | `"Inter Variable", "Inter", system-ui, -apple-system, "Segoe UI",` |
| `--vy-font-mono` | `"JetBrains Mono", ui-monospace, "SF Mono", "Cascadia Mono", Consc` |
| `--vy-text-2xs` | `12px` |
| `--vy-text-xs` | `13px` |
| `--vy-text-sm` | `14px` |
| `--vy-text-base` | `15px` |
| `--vy-text-md` | `16px` |
| `--vy-text-lg` | `18px` |
| `--vy-text-xl` | `20px` |
| `--vy-text-2xl` | `24px` |
| `--vy-text-3xl` | `28px` |
| `--vy-weight-regular` | `400` |
| `--vy-weight-medium` | `500` |
| `--vy-weight-semibold` | `600` |
| `--vy-weight-bold` | `650` |
| `--vy-weight-heavy` | `700` |

**The scale shifts on a touch device.** Under `@media (pointer: coarse)` the body range moves up by a constant **+2px** — 12, 13, 14, 15, 16, 18 instead of 10, 11, 12, 13, 14, 16 — which keeps every relationship intact and lifts the smallest step to a readable 12. Display sizes ( `xl` and up) do not move: they are already comfortable, and growing a heading on a 375px screen buys nothing and costs a wrap.

## Casing — three levels, drawn differently on purpose

The customer's UI Standard of 14 October: buttons, links, tabs, form and dialog titles, section and card titles, **field labels**, column headers, filter labels, menu items and status pills take **Title Case**, with prepositions (for, to, of, by, as, in, on, at, from, with, via, per, into) lowercase unless they lead. Prose is sentence case.

| Level | Case | Size / weight | Example |
|---|---|---|---|
| Card header | Title Case | 14px / 600 | `Select Action` · `Continue from Drafts` |
| Divider inside a card | **UPPERCASE**, .05em tracking | 12px / 600 | `QUOTE STRATEGY` · `BOM OPTIONS` |
| Field label | Title Case | 12px / 600 | `Customer` · `Qty to Quote` · `Markup (%)` |

# B3 · Spacing

## B3.1 The scale

| Token | Value |
| --- | --- |
| `--vy-space-1` | `2px` |
| `--vy-space-2` | `4px` |
| `--vy-space-3` | `6px` |
| `--vy-space-4` | `8px` |
| `--vy-space-5` | `10px` |
| `--vy-space-6` | `12px` |
| `--vy-space-7` | `14px` |
| `--vy-space-8` | `16px` |
| `--vy-space-9` | `20px` |
| `--vy-space-10` | `24px` |
| `--vy-space-11` | `28px` |
| `--vy-space-12` | `32px` |
| `--vy-space-14` | `40px` |

## B3.2 Which gap goes between two controls

The scale does not say which step to pick, so the system states it. The tiers are what carry **grouping**, which is why they are not collapsed into one value:

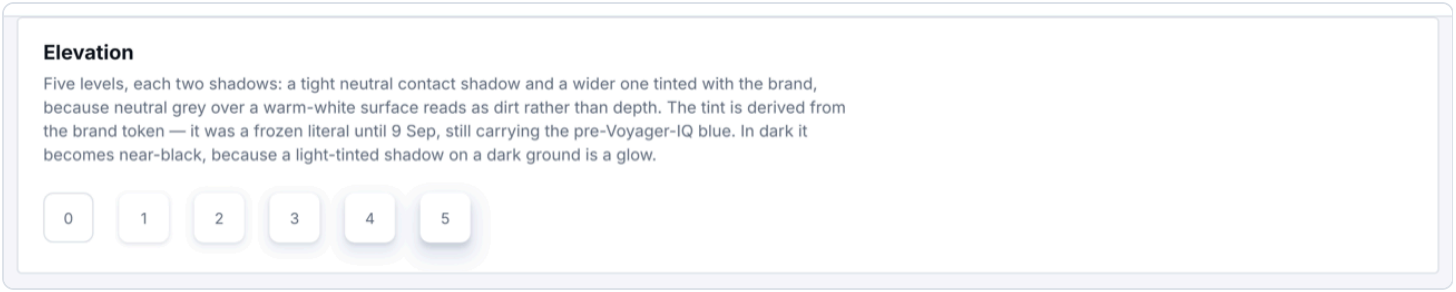
| Gap | Between | Examples |
|---|---|---|
| **2px** | Buttons in a cluster that reads as **one** control | dialog titlebar, pager arrows |
| **4px** | Icon buttons attached to the control they act on | a Select with Preview and Download |
| **8px** | **Two adjacent actions** — the default | `.vy-page-actions`, `.vy-run-nav-actions`, `.vy-cost-actions` |
| **12px** | Items that each hold a control **and** a label | filter checkboxes, which are 8px from their own words |
| **16px** | **Groups** on one row | toolbar filters / search / Columns |

Two buttons side by side are 8px apart everywhere. A bigger gap means a real boundary, not a different author.

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## B4 · Shape, elevation, depth

Four radii, three shadows. Elevation is for things that **float** — a dialog, a menu, a toast. A card on a page gets a border, not a shadow.



| Token | Value | Dark |
|---|---|---|
| `--vy-radius-sm` | `3px` | *(same)* |
| `--vy-radius-md` | `4px` | *(same)* |
| `--vy-radius-lg` | `6px` | *(same)* |
| `--vy-radius-pill` | `999px` | *(same)* |
| `--vy-shadow-sm` | `0 1px 2px rgba(19, 24, 32, .06)` | *(same)* |
| `--vy-shadow-md` | `0 2px 8px rgba(19, 24, 32, .10)` | *(same)* |
| `--vy-shadow-lg` | `0 8px 28px rgba(19, 24, 32, .14)` | *(same)* |
| `--vy-overlay-scrim` | `rgba(19, 24, 32, .45)` | `rgba(0, 0, 0, .62)` |

## Z-index — one flat list, so two overlays can never argue

| Token | Value |
|---|---|
| `--vy-shell-sidebar-w` | `248px` |
| `--vy-shell-sidebar-w-collapsed` | `56px` |
| `--vy-shell-topbar-h` | `52px` |
| `--vy-z-drawer` | `10020` |
| `--vy-z-dialog` | `10040` |
| `--vy-z-menu` | `10070` |
| `--vy-z-palette` | `10075` |
| `--vy-z-toast` | `12500` |
| `--vy-z-skip` | `12600` |
| `--vy-dialog-md` | `640px` |
| `--vy-dialog-lg` | `960px` |
| `--vy-dialog-xl` | `1180px` |

# B5 · Sizing

## B5.1 Control heights

Every control sits on **Kendo's own size scale**, configured rather than overridden — `_kendo-size.scss` sets the variables and the components inherit them. Forcing a height on a Kendo control instead is how a 29px select ends up beside a 34px input.

| Token | Value |
|---|---|
| `--vy-row-h-compact` | `28px` |
| `--vy-row-h-comfortable` | `40px` |
| `--vy-row-h-relaxed` | `48px` |
| `--vy-target-min` | `44px` |
| `--vy-control-sm` | `28px` |
| `--vy-control-row` | `26px` |
| `--vy-control-md` | `34px` |
| `--vy-control-dense` | `32px` |
| `--vy-icon-sm` | `16px` |

| Token | Where it is used |
|---|---|
| `--vy-control-md` **34px** | Form inputs, selects, buttons — the default |
| `--vy-control-dense` **32px** | Toolbar controls: search, `size="sm"` buttons |
| `--vy-control-sm` **28px** | Compact density |
| `--vy-control-row` **26px** | An icon button **inside a grid row** — Kendo's `xs` step |
| `--vy-target-min` **44px** | A floor, under `pointer: coarse` **only** |

> **44px is a touch rule, not a desktop one.** WCAG 2.5.5 is about a pointer target; applying it to a 27-field form on a mouse-driven screen adds ~270px of scrolling for nothing.

## B5.2 Row density — a user preference

| Token | Height | Setting |
|---|---|---|
| `--vy-row-h-compact` | 28px | Compact |
| `--vy-row-h-comfortable` | 40px | Comfortable *(default)* |
| `--vy-row-h-relaxed` | 48px | Relaxed |

## B5.3 Column roles — a width is chosen once, by role

A grid column declares a **role**, not a pixel width.

**Column roles**

The rule that replaces uniform 108px columns. Width is a consequence of what the column holds — and a column empty in more than half its rows is hidden by default, with the reason stated in the chooser.

| ROLE | WIDTH | RULE |
|---|---|---|
| ident | 240px | The primary identifier. Monospaced, tabular, never truncated — it is how people refer to the record out loud and in email. |
| text | 280px | The only role allowed to truncate. Fixed at 280px with the full value in a tooltip; the grid scrolls rather than squeezing every column. |
| code | 96px | Short enumerations. Width is set to the longest member of the enum, so it never clips and never over-reserves. |
| number | 104px | Right-aligned with tabular figures so magnitudes line up and a misplaced digit is visible. |
| money | 124px | Right-aligned, currency-aware, always two decimals. |
| date | 150px | Sized to the full rendered format. A truncated date is worse than no date — production truncates 100% of them. |
| datetime | 190px | — |
| status | 128px | One badge from the shared status tokens. Never free text. |
| priority | 104px | — |

| Role | Width | What it holds |
|---|---|---|
| `ident` | 240px | Part numbers, MPNs — must not truncate |
| `text` | 280px | Descriptions and other prose |
| `code` | 96px | Short codes, revisions |
| `number` | 104px | Quantities |
| `money` | 124px | Prices and amounts |
| `date` | 150px | A day |
| `datetime` | 190px | A moment — `09/09/2026 16:35:29` |
| `status` | 128px | A status pill |
| `priority` | 104px | A dot and a word |

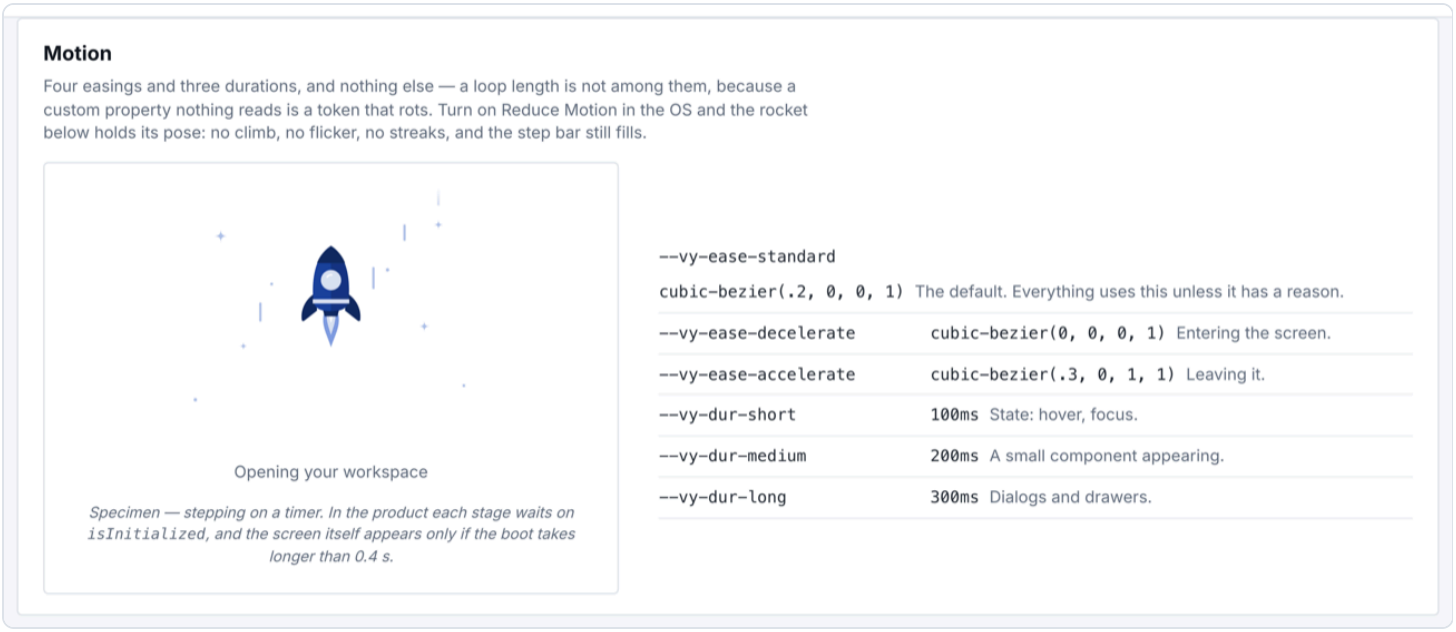
## B5.4 Dialogs and shell

| Size | Width | Opens maximised? | For |
|---|---|---|---|
| `md` | 640px | No | A question: a confirmation, a short form |
| `lg` | 960px | Yes *(overridable)* | Two blocks and a short table |
| `xl` | 1180px | Yes | A work surface: a 21-column grid, a 4-step wizard |

| Shell token | Value |
|---|---|
| `--vy-shell-sidebar-w` | 248px |
| `--vy-shell-sidebar-w-collapsed` | 56px |
| `--vy-shell-topbar-h` | 52px |

# B6 · Motion

Four easings and three durations, and nothing else. There is deliberately **no loop-length token**: a custom property nothing reads is a token that rots.




| Token | Value | Use |
|---|---|---|
| `--vy-ease-standard` | `cubic-bezier(.2, 0, 0, 1)` | The default. Everything, unless it has a reason |
| `--vy-ease-emphasised` | `cubic-bezier(.2, 0, 0, 1)` | A moment that should be noticed. **Currently identical to** `standard` — named separately so the two can diverge without touching call sites |
| `--vy-ease-decelerate` | `cubic-bezier(0, 0, 0, 1)` | Something arriving |
| `--vy-ease-accelerate` | `cubic-bezier(.3, 0, 1, 1)` | Something leaving |
| `--vy-dur-short` | **100ms** | State changes: hover, focus |
| `--vy-dur-medium` | **200ms** | Small components entering |
| `--vy-dur-long` | **300ms** | Dialogs, drawers |

## The loading clock

| Rule | Value |
|---|---|
| Show the boot screen only if the boot takes longer than | **400ms** |
| Exit — the craft leaves the top of the frame | **400ms** |
| "Still climbing…" appears after | **5s** |
| Minimum display time once shown | **none** |

Holding a loading screen so it does not flash spends the user's time to make our indicator look better. Not showing one at all for a boot that beats the clock costs them nothing. The two rules are the two halves of one answer.

### Reduced motion is a state, not a switch

Under `prefers-reduced-motion: reduce` the rocket **holds its pose** — no climb, no flicker, no streaks — and the step bar still fills. A welding arm with its animation removed is an arm with no sparks and an empty checklist, which says something different from what the state means, so the things that *fade in* are left on.

---

## B7 · Responsive

Breakpoints are raw pixels because `@media` cannot read custom properties — a documented exception to the no-magic-numbers rule.

| Width | What changes |
|---|---|
| ≤ **1180px** | The 248px sidebar becomes a 56px icon rail |
| ≤ **980px** | Two-column regions stack |
| ≤ **820px** | The sidebar leaves the flow; a menu button reveals it |
| `pointer: coarse` | Type scale +2px on the body range; 44px target floor |

The system runs on shop-floor terminals that are often 1366×768 or narrower. Below about 1100px a 248px sidebar stops being orientation and starts being the thing in the way — measured at 543px it was taking **46% of the viewport**.

---

# Part C — Assets

## C1 · Icons

One set. **20×20 viewBox, stroke only, no fill, one stroke weight.** A heavier line at a smaller size is how an icon set stops looking like one set.

An unknown name renders **nothing**, with the name left in the DOM, rather than falling back to a document glyph that hides the bug.

| | |
|---|---|
| Default size | 16px · `--vy-icon-sm` |
| In a grid row | 15px inside a 26px button |
| Colour | `currentColor` — the icon takes the text colour around it |
| Decorative | `aria-hidden`, and the state it marks gets words of its own |

**The 45 names:** `task · chat · log · quote · order · return · customer · requisition · supplier · factory · simulate · bom · transfer · adjust · pcb · tool · machine · invoice · journal · lock · people · template · job · plug · menu · globe · metadata · queue · doc · home · sell · buy · parts · stock · make · finance · insight · settings · chevron-down · chevron-right · eye · download · clip · trash · check`

Two directional glyphs, each with one job: **down** opens a menu below itself, **right** says "this opens somewhere else".

## C2 · Illustrations

**One drawing, three states.** The product has exactly one illustrated character — a rocket — and it appears in three situations. That is deliberate: a reader who has met this craft on the way in should not have to learn a second picture to understand that it crashed.

Component: `src/ui/RocketLaunch.tsx` · one inline SVG, `viewBox="0 0 240 180"` · `data-state` selects the scene.

| | |
|---|---|
| Rendered width | `clamp(240px, 18vw, 280px)` |
| Colour | Ten role tokens, B1.7 — never hex |
| Motion | CSS only. The illustration owns pixels and no state, so it can stop in a background tab and nothing is lost |
| Ids | Generated with `useId` — two instances on one page would otherwise share a clip path |
| Inline, not `<img>` | An `<img>` is a second request on the one screen whose job is to cover the first one, and it is its own document, where `var(--vy-brand)` resolves against nothing |

## C2.1 `launching` — the boot screen

The craft holds still in the middle of the frame and **the world moves**: the planet sinks a notch per boot stage, streaks run down past it. Nothing about a rocket drawn nose-up says "climbing" on its own.

| | |
|---|---|
| Where | `BootGate` after sign-in; the `Suspense` fallback of a lazy route |
| Animation | Lift-off 0.7s · bob 1.6s · flame flicker 0.18s · streaks 0.8–1.3s |
| Reduced motion | No climb, no flicker, no streaks. The step bar still fills |



Shown with its label and five-segment bar, which is what the boot screen actually is:



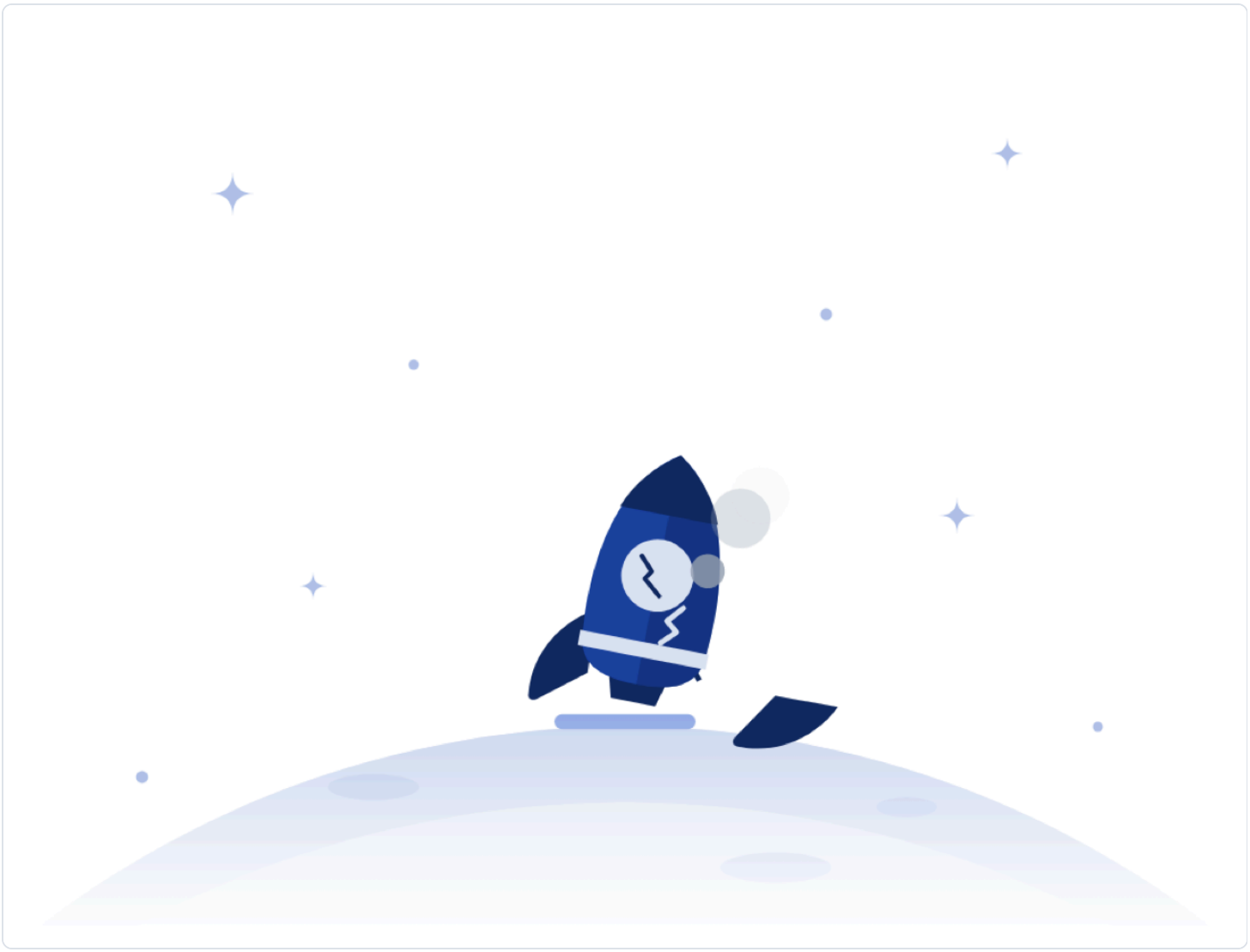
Restoring your settings

*Specimen — stepping on a timer. In the product each stage waits on `CHANGE_FORMAT_CONFIG`, and the screen itself appears only if the boot takes longer than 0.4 s.*

## C2.2 `error` — the screen stopped loading

The same craft, back on the pad: tilted, cracked, one fin lying beside it, smoking. The failure is legible before the sentence under it is read.

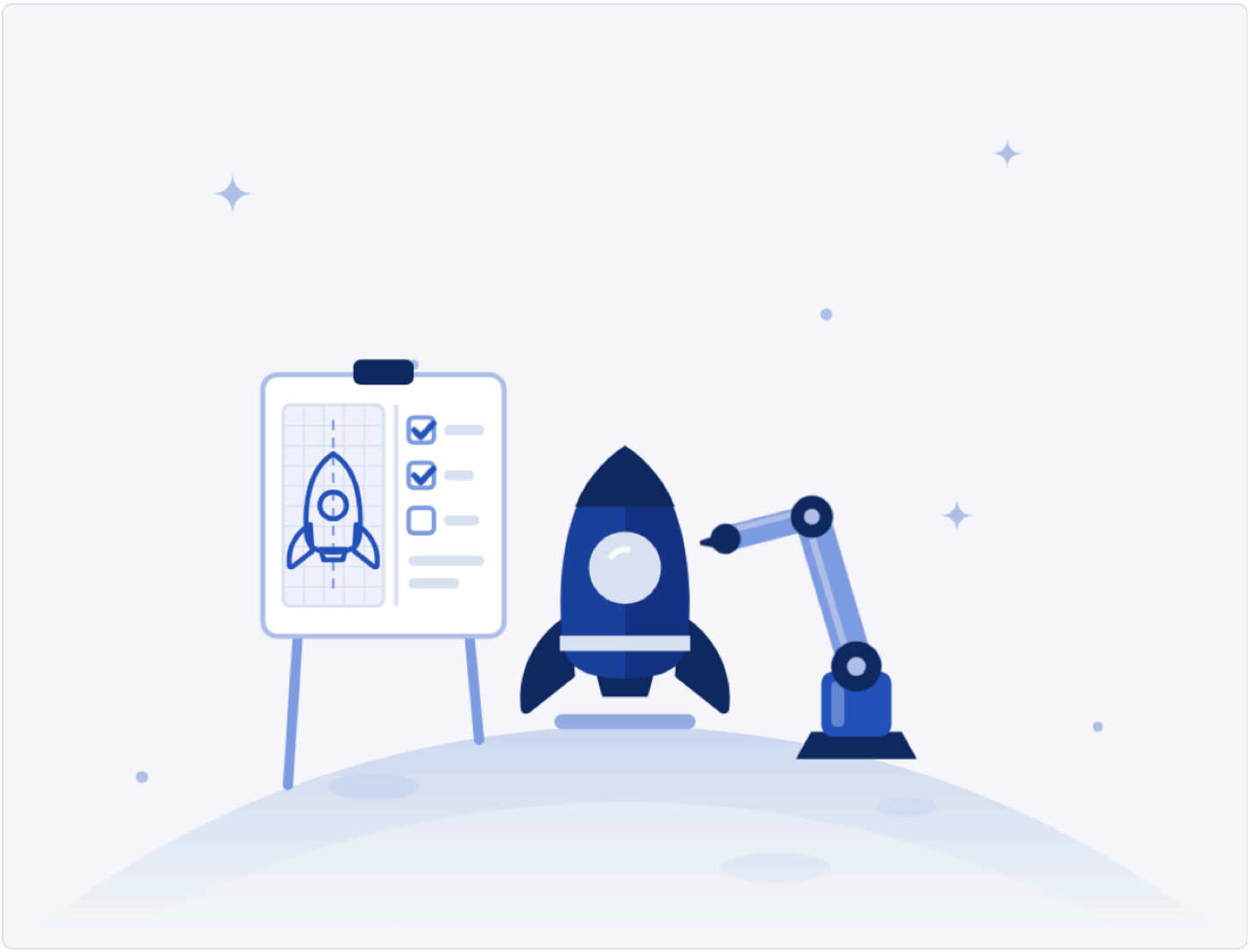
| | |
|---|---|
| Where | `ErrorState`, which `ErrorBoundary` renders for any route that throws |
| Animation | Three smoke puffs, 2.4s, staggered |
| Reduced motion | Puffs held at 0.6 opacity — the smoke stays, the movement goes |



## C2.3 `building` — still on the launch pad

The craft **whole** on the pad, a robot arm welding it, a design doc on an easel whose checklist ticks off. Work in progress, not wreckage — and the difference has to be visible before the heading is read.

| | |
|---|---|
| Where | `Placeholder`, which answers every route not yet built (~70 of 91) |
| Animation | Arm 4s · sparks only while the arm is holding · checklist 6s |
| Reduced motion | Sparks and the first two ticks held **on**, so the drawing still says "being built" |

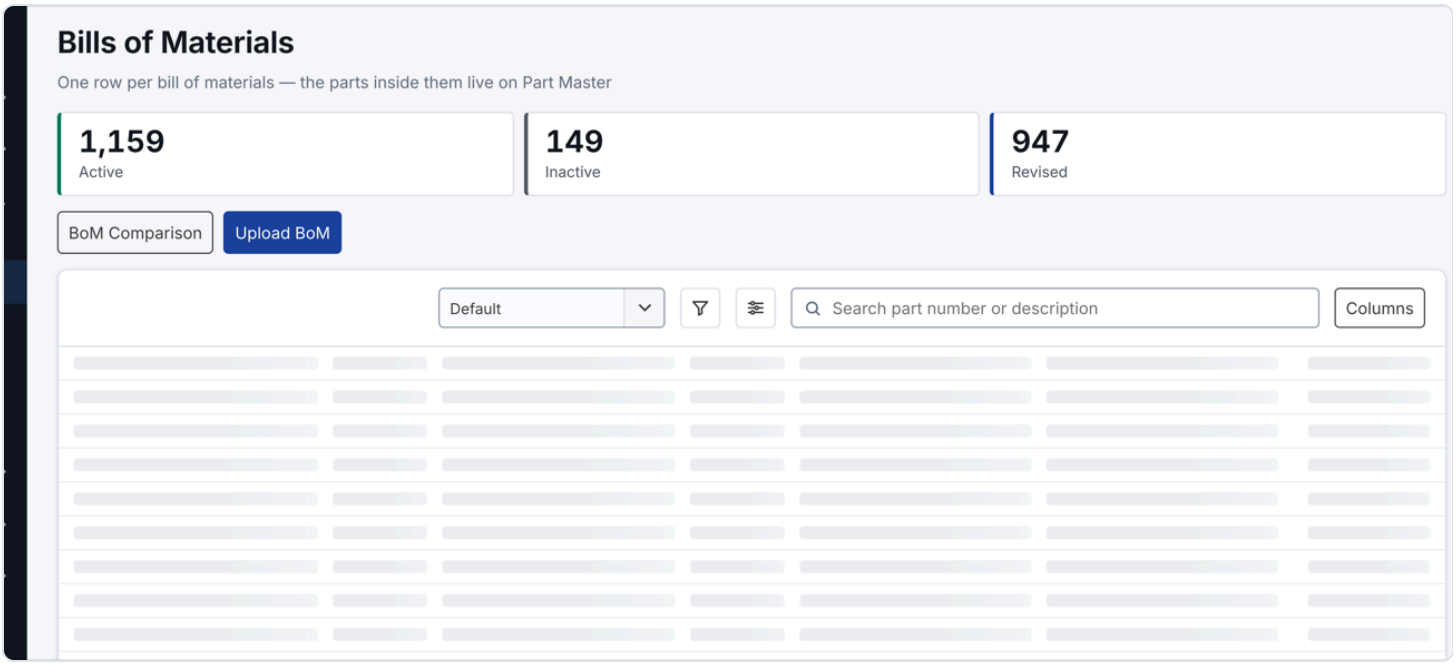


## C2.4 Skeletons — the other kind of waiting

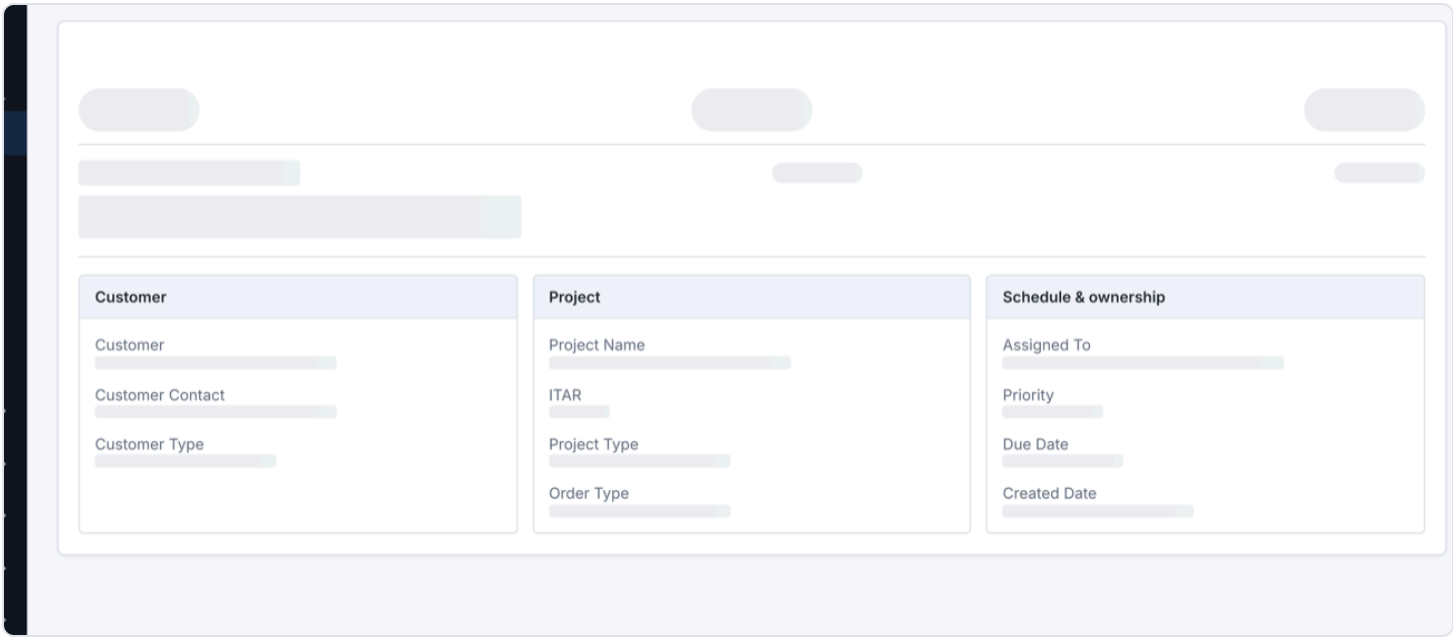
A skeleton answers *"what will this look like?"*. Once the user has already seen rows that question is answered, so a re-price or a second page **dims what is there** and puts an indicator over it instead (`BusyOverlay`).

One shimmer for both: `--vy-surface-sunken` → `--vy-surface-app` → `--vy-surface-sunken`, 200% background, 1.4s.

**Grid skeleton** — the real column roles, so the bars land where the data will:




**Record skeleton** — the real field labels, because they are known before the data is. Measured down to 1px of shift when the record arrives:




## C2.5 Empty states

No illustration, on purpose. An empty state is a sentence and an action, and a picture there delays both.




## C2.6 Exporting and regenerating

| | |
|---|---|
| Live source | `src/ui/RocketLaunch.tsx` |
| Static SVGs | `docs/assets/rocket-journey/*.svg` — the **retired** horizontal set, kept because the 16 Oct comparison document uses them. Not in the product |
| These images | Captured from the running app at 2×; the scripts are in `scripts/` |

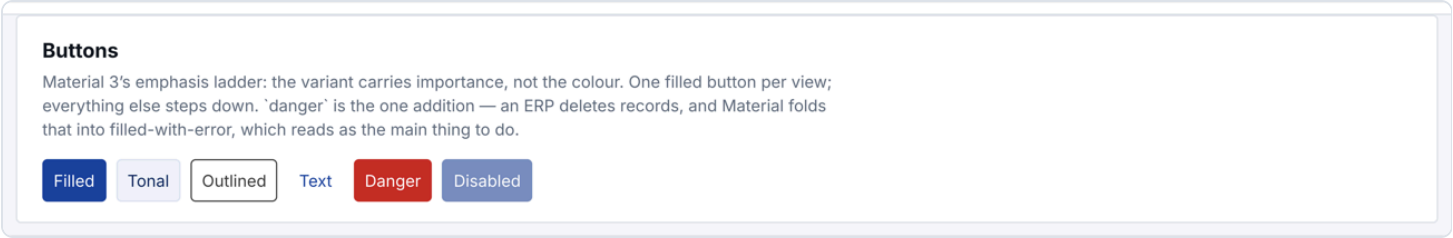
> **If an illustration needs to leave the codebase** — for a slide, a PDF — export the inline SVG and replace the `var(--vy-rocket-*)` references with the resolved hex from B1.7. Outside the app there is no token layer to resolve them.

---

# Part D — Components and patterns

`src/ui` holds 33 files: the primitives, plus the icon set, the preference store and three hooks. Each wrapper has a reason, and the rule that produced them is the same every time — **the live system already ships KendoReact, so the question is never "what should this look like" but "what does Kendo already do, and what must we add".**

## D1 · Buttons



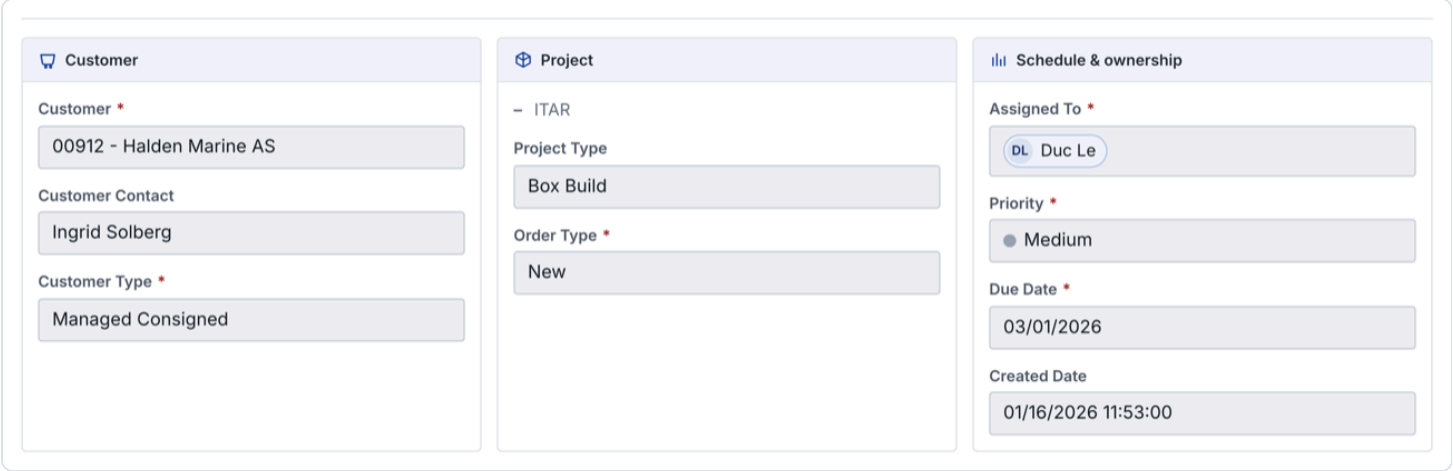

| Variant | When |
|---|---|
| `filled` | **One per view.** The most likely next step |
| `outlined` | The default for everything else |
| `tonal` | A secondary action that still needs weight |
| `text` | Tertiary, inside dense rows |
| `danger` | Destructive — and it goes behind **More** ▾ |

**Placement:** all buttons on the **left** of a toolbar; a smart-button group or step indicator on the right; at most four visible, the rest behind `More ▾` — and the label is always the word *More* with a chevron, never `⋯`.

**Cancel only in dialogs.** In edit mode the word is **Discard Changes**.

`busy` **is not decoration.** The indicator says something is happening; the **disable** is what stops a second click spending a second marketplace request.

## D2 · Fields




**Two backgrounds, not three** — the 14 October instruction: *white means you can type in it.* A read-only value is a box with a sunken fill, a 1px border and a padlock, so read and edit modes are the same shape and the layout does not move between them.

| Part | Rule |
|---|---|
| Label | **Above** the value, never beside it (13 Oct). Title Case, 12px, 600 |
| Required | A **red asterisk** right after the label, in view **and** edit mode |
| Optional | No mark. Never the word "(optional)" |
| Helper | Below the field, italic, ≤ the label's size (22 Oct) |
| Error | Red text below the field, **on blur** — not on every keystroke |
| Hint | A **sibling** of the label, never inside it, or it joins the accessible name |

## D3 · Grids

| Rule | |
|---|---|
| Column chooser | On every grid with **3+ columns** |
| Frozen block | Declared as `locked`, so the chooser and the freeze cannot disagree |
| Zebra | By **record**, not by row — a record spanning two AML rows is one band |
| Totals | A grid the user sums by eye gets a totals row |
| Empty | A real empty state that says what to do, never "No records available" |
| Width | By role (B5.3); a deviation carries a `widthNote` |

## D4 · Dialogs

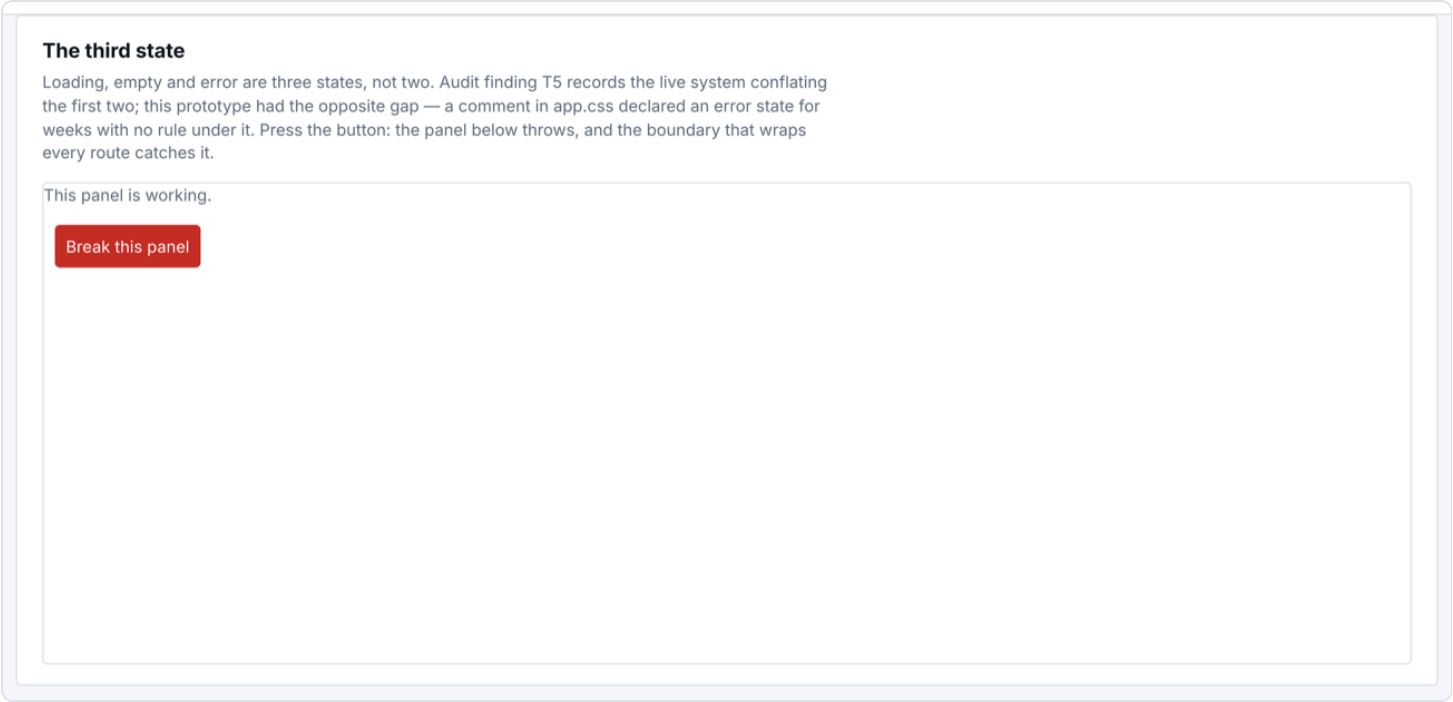
Sizes are in B5.4. Three further rules:

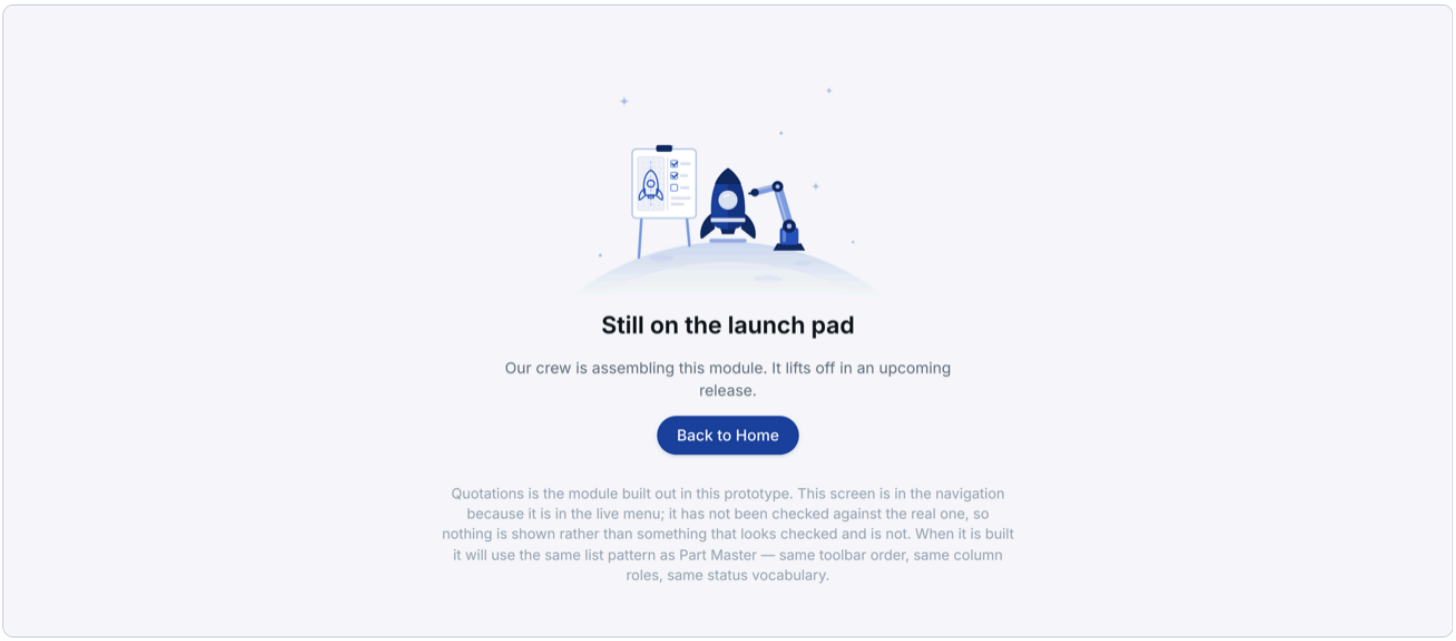
1. **Actions top-left** on a work surface, in a footer on a question. The way out goes on the left wherever the bar sits.
2. `lg` **and** `xl` **open maximised** — asked for at the 13 October review. `md` never does: a confirmation filling a 27-inch screen is a worse answer than the one it replaced.
3. **Escape dismisses the innermost thing**, which the Select had to be taught by hand — Kendo's own handling closed the dialog from an open dropdown.

## D5 · The four screen states

Loading, empty, error and not-yet-built are four different things and look it. Getting these wrong is the most common way a screen lies to its reader.

| State | What the user sees | Actions | Illustration |
|---|---|---|---|
| **Loading** | Rocket climbing, planet sinking, five-segment bar | none | C2.1 |
| **Empty** | A sentence saying what would fill it | the action that fills it | none |
| **Error** | Rocket on the pad: tilted, cracked, one fin off, smoking | Reload · Back to Home · Send a support ticket · Error ID + Copy | C2.2 |
| **Under development** | Rocket whole on the pad, arm welding, plan ticking off | Back to Home. **No Reload — nothing is broken** | C2.3 |






**The error state's three slots** are the three questions a person actually has: *what failed · what it means for me · what I can do next.* Only the first is generic.

**The error id is minted where the error is caught**, not where it is drawn — minted in render, pressing Copy would hand the user a different number for the same failure.

## D6 · Words

| | |
|---|---|
| Casing | B2 — Title Case for labels, sentence case for prose |
| A control says what happens | `Publish`, then a toast saying `Published` |
| An error says what to do | Not "Something went wrong" |
| Nothing is "not implemented" silently | A toast says what it *would* do |
| A derived figure says it is derived | *"Derived from this run — the live system takes them from the customer's purchase history"* |
| A session-only change says so | *"Added in this session… nothing here is saved"* |

# Part E — Quality

## E1 · Accessibility

Measured, not asserted. The full sweep is in `contrast-sweep.md`; the keyboard work is in `focus-order.md`.

| Requirement | Threshold | Status |
|---|---|---|
| Text contrast — WCAG 1.4.3 | 4.5:1 normal · 3:1 large (≥24px, or ≥18.66px at ≥700) | **78 combinations measured, 0 failures** |
| Non-text contrast — WCAG 1.4.11 | 3:1 for control boundaries | Three controls failed at 1.30 / 1.64 / 2.55 · fixed by `--vy-border-control` at **3.06:1** |
| Target size — WCAG 2.5.5 | 44×44 | Enforced under `pointer: coarse` |
| Reduced motion | respected | B6 |

**Why the contrast harness can be trusted:** colours are read through a **canvas pixel** rather than parsed from strings, because Kendo emits `oklch()`, which no regex over `getComputedStyle` survives. Backgrounds are composited up the ancestor chain. Before the results were believed, a `#bbbbbb`-on-white element was planted in the page and caught at **1.92:1** — a sweep that reports no failures is only worth reading if it can prove it would have found one.

Four rules that are easy to get wrong and are already settled:

- A **status is a word and a colour**, never a colour alone.
- A **hint is a sibling of the label**, never inside it.
- `role="alert"` **for an error,** `role="status"` **for a loading state.** One interrupts, the other does not.
- A **decorative glyph is** `aria-hidden` and the state it marks gets words of its own — a padlock plus a screen-reader-only "read-only".

## E2 · KendoReact

The live system already ships KendoReact 16. The bridge is three files:

| File | Job |
| --- | --- |
| `kendo-subset.scss` | Compiles only the Kendo components actually used |
| `_kendo-size.scss` | **Configures** Kendo's size scale — does not override it |
| `kendo-bridge.css` | Maps Kendo's variables onto ours, and fixes what cannot be configured |

**Specificity is the trap.** `.k-button.k-button-solid.k-button-base.k-focus` weighs (0,4,0), so a one-class rule of ours loses silently. The bridge prefixes with `html[data-kendo-bridged]` rather than repeating a class five times.

Two Kendo behaviours worth knowing before debugging a component:

- **Kendo disables with a `.k-disabled` CLASS, not the `:disabled` attribute.** A plain `<input disabled class="k-radio">` renders identically to an enabled one.
- **KendoReact 16 locks grid columns with sticky cells in one table** ( `position: sticky` ), not with two panes.

---

# Part F — Handover

# F1 · File map

| Path | Contents |
|---|---|
| `src/theme/tokens.css` | **Every token.** Light, plus the two dark blocks |
| `src/theme/tokens.ts` | The values TypeScript needs: column widths, dialog widths |
| `src/theme/md3.css` | Motion, focus ring, the Material 3 structure we kept |
| `src/theme/base.css` | Reset and element defaults |
| `src/theme/components.css` | Component rules |
| `src/theme/app.css` | Screen and layout rules |
| `src/theme/responsive.css` | Every breakpoint, in one file |
| `src/theme/kendo-*.{css,scss}` | The Kendo bridge |
| `src/ui/` | The primitives |
| `src/ui/RocketLaunch.tsx` | The illustration, all three states |
| `/design-system` | The living specimen page |
| `docs/assets/design-system/` | The images in this document |

# F2 · How to change something

### A colour token

1. Add or change it in **all three** blocks of `tokens.css` (B1.8).
2. If it lands on text or a control boundary, re-run the contrast sweep.
3. `npm run check:fast` — lint, stylesheet check, typecheck, build.

### A component

1. Check whether Kendo already does it (E2).
2. Build it in `src/ui`, referencing semantic tokens only.
3. Add a specimen to `/design-system` if it has states worth seeing.

4. Cover every state: default, hover, focus, active, disabled, busy, error.

**A new screen**

1. Which of the four states (D5) can it be in? Build all of them.
2. Column roles, not pixel widths.
3. Check it at 1180, 980 and 820, in both themes.

**What CI runs on every push:** `render:check` · `touch:check` · `mobile:check` · `focus:check` · `hover:check`.

## F3 · Governance

| | |
|---|---|
| Owner | The UX revamp working copy, `ll-poc` |
| Where decisions are recorded | A commit message, and a dated note in `docs/` for anything a reader would otherwise have to re-derive |
| Precedence when two sources disagree | `precedence.md` — customer requirement, then the live system, then judgement, and judgement must be labelled |
| Versioning | `v1.0` is this handover. A token or rule change is a minor version; a change that requires screens to be rebuilt is a major one |

**Three rules that keep this document true:**

1. **A magic number needs a note** saying what was measured.
2. **A comment states what was measured, not what was intended.** Three corrections in this project came from a comment describing behaviour the code did not have.
3. **The living page beats the document.** `/design-system` renders the real components; if the two disagree, the page is right and this file is stale.

## F4 · Known gaps

An honest list, because a handover claiming completeness is one nobody checks.

| | Status |
|---|---|
| **Label text casing** ( `Reload this screen` → `Reload This Screen` ) | Half done. Uppercase is gone from labels; rewriting the **words** collides with the customer's own prototypes, which use sentence case. Awaiting her answer |
| **Sale Type list** on Other Quoted Cost | Eight values: two hers, six ours. Awaiting the metadata list |
| **Support ticket destination** | The dialog is complete; the send is faked. One POST away |
| **Row merging on step 3** | Her prototype merges cells; ours bands by record. Trade-off documented, awaiting a decision |
| **Select Action order** | Her prototype and her UI Standard disagree |
| `Customer Special Need` vs `Customer specific needs` | Her prototype and the live system disagree |
| `--vy-ease-emphasised` | Identical to `standard`. Either give it its own curve or retire it |

## F5 · What a handover should contain

The shape of this document was checked against current practice rather than invented. The recurring list across published checklists is: **design tokens** with their values · **component specs** with every state and variant · **usage rules** with do and don't · **accessibility** with the WCAG thresholds named · **code examples** · **responsive** behaviour with the breakpoints stated · **motion** with duration, easing and a reduced-motion alternative · and **governance** — ownership, a contribution process, versioning.

Several sources also name the one failure mode this file is most exposed to: documentation drifting away from the live code once the team moves on. That is why F3 makes the specimen page the tie-breaker rather than this document.

Consulted 3 November 2026: Magic Patterns · Figr checklist · Supernova · Docsie · Eleken · UX Planet. These are vendor blogs and community checklists, not a standard; they agree on the core set and differ on depth.

# F6 · Source index

| Document | What it settles |
|---|---|
| `precedence.md` | Which source wins when two disagree |
| `contrast-sweep.md` | Every measured contrast ratio, and the method |
| `focus-order.md` | Keyboard order and the focus ring |
| `modal-patterns.md` | Dialog sizes, action placement, dismissal |
| `table-patterns.md` | Grid behaviour, column roles, density |
| `motion-loading.md` | Every loading and transition decision |
| `kendo-control-scale-18oct.md` | The 33-rule Kendo size audit |
| `ui-standard-review-16oct.md` | Our review of the customer's UI Standard |
| `working-agreement.md` | Engineering rules, including the magic-number exceptions |
| `con-thieu-01nov.md` | What is still outstanding against the customer's own material |

*Captured from the running build at 2× on 3 November 2026. Images live in `docs/assets/design-system/` and can be regenerated from `/design-system` and the product screens they came from.*